6	ai	The *magnetic flux density at a point* is defined as the force acting per unit current per unit length on a current carrying conductor	M1
		when the conductor is placed at right angles to the magnetic field.	B1
	aii	As AB moves from P towards Q, magnetic flux linkage over the area ABCD enclosed by the frame increases resulting in an induced e.m.f. generated in the frame by Faraday's law.	B1
		Since the frame is a conductor, induced current flows. By Lenz's Law, the induced current flows in an anticlockwise direction.	B1
		This results in a magnetic force that acts on AB towards the left which oppose motion. Hence the frame slows down.	B1
		Alternative:	
		As AB moves from P towards Q, magnetic flux linkage over the area ABCD enclosed by the frame increases resulting in an induced e.m.f. generated in the frame by faraday's law.	B1
		Frame is a closed circuit so induced current will flow in the frame. By conservation of energy, KE of the frame is converted to heat energy.	B1
		Energy is loss as $P = I^2R$. Hence frame slows down	B1
	aiii	As AB enters the field, the increase in magnetic flux (linkage) $\Phi = BA = B(wx)$ where *x* is the distance AB has moved past P.	B1
		Hence, the magnitude of induced emf is given by $E = \frac{d\Phi}{dt} = Bw\frac{dx}{dt} = Bwv$	B1
		Induced current that flows in the frame, $I = \frac{Bwv}{R}$	B1
		Magnetic force which acts on AB is in opposite direction to motion is the braking force which slows the frame.	B1
		Braking force is thus $F = BIw = \frac{B^2w^2v}{R}$	A0
	bii	Distance **2*L*** = Area under *v* – *t* graph	
		= ½ (26 + 16 x 2 + 10 x 2 + 6 x 2 + 3.5)(5)	C1
		= 234 m	
		PQ = *L* = 117 m, Accuracy: L = 105 to 129 m	A1
	biii	Braking force only acts on the train when there is a changing magnetic flux through the frame. There is no braking force on the train when the whole frame is within the magnetic field.	M1
		Increasing the length PQ does not change the exit speed.	A1

7(a)(i) P